OS & SP

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SKILL – 21

1.Skill21.c

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GNU nano 8.7.1 skill21.c *

```c
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <sys/wait.h>
#include <errno.h>

#define MAX_INPUT 256
#define MAX_ARGS 64

/* Display error message */
void print_error(const char *message)
{
    fprintf(stderr, "ERROR: %s\n", message);
}

/* Parse command */
int parse_command(char *input, char *args[])
{
    int argc = 0;
    char *token = strtok(input, " ");

    while (token != NULL && argc < MAX_ARGS - 1) {
        args[argc++] = token;
        token = strtok(NULL, " ");
    }

    args[argc] = NULL;
    return argc;
}

/* Execute command */
int execute_command(char *args[])
{
    pid_t pid = fork();

    if (pid < 0) {
        perror("fork");
        return -1;
    }

    /* Child process */
    if (pid == 0) {
        execvp(args[0], args);

        /* Executes only if execvp fails */
        fprintf(stderr,
                "ERROR: Cannot execute '%s': %s\n",
                args[0], strerror(errno));

        exit(127);
    }

    /* Parent process */
    else {
        int status;

        if (waitpid(pid, &status, 0) < 0) {
```

```c
        if (strcmp(input, "exit") == 0) {
            printf("Exiting safely...\n");
            break;
        }

        /* Help */
        if (strcmp(input, "help") == 0) {
            printf("\nAvailable commands:\n");
            printf(" help - display help\n");
            printf(" exit - exit shell\n");
            printf(" Any Linux command\n\n");
            continue;
        }

        /* Copy input because strtok modifies it */
        char command_copy[MAX_INPUT];
        strcpy(command_copy, input);

        /* Argument array */
        char *args[MAX_ARGS];

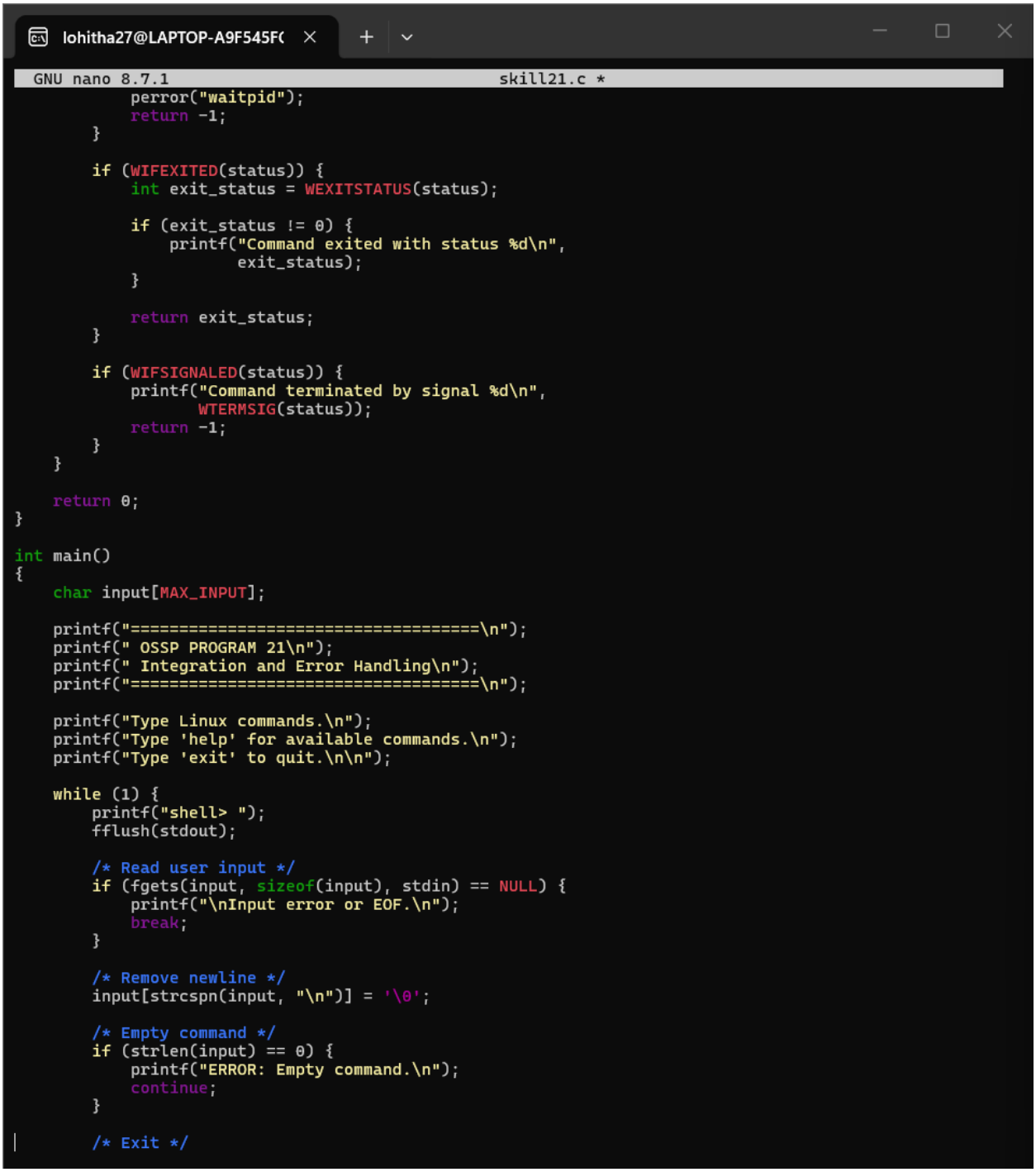
        /* Parse command */
        int argc = parse_command(command_copy, args);

        /* Validate parser output */
        if (argc == 0) {
            print_error("Invalid command syntax.");
            continue;
        }

        /* Execute command */
        int result = execute_command(args);

        /* Runtime error handling */
        if (result == -1) {
            print_error("Command execution failed.");
        }
    }

    printf("Program 21 terminated.\n");
    return 0;
}
```


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GNU nano 8.7.1 skill21.c *
perror("waitpid");
return -1;
}
if (WIFEXITED(status)) {
int exit_status = WEXITSTATUS(status);
if (exit_status != 0) {
printf("Command exited with status %d\n",
exit_status);
}
return exit_status;
}
if (WIFSIGNALED(status)) {
printf("Command terminated by signal %d\n",
WTERMSIG(status));
return -1;
}
}
return 0;
}
int main()
{
char input[MAX_INPUT];
printf("====================================\n");
printf(" OSSP PROGRAM 21\n");
printf(" Integration and Error Handling\n");
printf("====================================\n");
printf("Type Linux commands.\n");
printf("Type 'help' for available commands.\n");
printf("Type 'exit' to quit.\n\n");
while (1) {
printf("shell> ");
fflush(stdout);
/* Read user input */
if (fgets(input, sizeof(input), stdin) == NULL) {
printf("\nInput error or EOF.\n");
break;
}
/* Remove newline */
input[strcspn(input, "\n")] = '\0';
/* Empty command */
if (strlen(input) == 0) {
printf("ERROR: Empty command.\n");
continue;
}
/* Exit */

OUTPUT :

```
Program 20 terminated.
lohitha27@LAPTOP-A9F545FG:~/SKILL$ nano skill21.c
lohitha27@LAPTOP-A9F545FG:~/SKILL$ gcc -Wall -Wextra -g skill21.c -o skill21
lohitha27@LAPTOP-A9F545FG:~/SKILL$ ./skill21
====================================
 OSSP PROGRAM 21
 Integration and Error Handling
====================================
Type Linux commands.
Type 'help' for available commands.
Type 'exit' to quit.

shell> pwd
/home/lohitha27/SKILL
shell> cal
    October 2026
Su Mo Tu We Th Fr Sa
             1  2  3
 4  5  6  7  8  9 10
11 12 13 14 15 16 17
18 19 20 21 22 23 24
25 26 27 28 29 30 31

shell> exit
Exiting safely...
Program 21 terminated.
lohitha27@LAPTOP-A9F545FG:~/SKILL$
```